

Permanent model of a curve

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Let (K, v_K) be a field. We first fix some notations and assumptions:

- $v_K : K \rightarrow \mathbb{R} \cup \{\infty\}$ a non-trivial discrete valuation, with respect to which K is complete,
- $\Gamma_K := v_K(K^\times)$ the value group,
- $\mathcal{O}_K \subseteq K$ the valuation ring,
- $\mathfrak{m}_K \subseteq K$ the maximal ideal,
- π_K the uniformizer of $\mathcal{O}_K$,
- $\mathbb{F}_K := \mathcal{O}_K/\mathfrak{m}_K$ the residue field, which we assume perfect with positive character,
- X will always denote a nice curve over K , unless otherwise specified, i.e. a smooth, projective and absolutely irreducible variety of dimension 1, and
- F_X the function field of X .

For a finite extension L/K , the valuation v_K will be uniquely extended to L and denoted as v_L . We will denote the ramification index with $e(v_L/v_K) = [\Gamma_{v_L} : \Gamma_{v_K}]$. The above notations can be applied to L by analogy. We will denote $X_L := X \otimes_K L$ the base change.

1 Model with reduced special fiber

Definition 1.1. Let X be a nice K -curve.

1. A *model* of X is a flat, proper and normal $\mathcal{O}_K$ -scheme $\mathcal{X}$ with generic fibre X , i.e. $\mathcal{X} \otimes_{\mathcal{O}_K} K = X$. We will denote $\mathcal{X}_s := \mathcal{X} \otimes_{\mathcal{O}_K} \mathbb{F}_K$ its special fiber.
2. A model X is *permanent* if $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is normal for all finite field extension L/K .

Remark 1.2. Since flatness and properness are preserved under base change, being permanent is equivalent to that $\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is a model of X_L .

Lemma 1.3. Let $\mathcal{X}$ be a flat and proper $\mathcal{O}_K$ -scheme such that $\mathcal{X} \otimes_{\mathcal{O}_K} K = X$. Denote $\mathcal{X}_s = \mathcal{X} \otimes_{\mathcal{O}_K} \mathbb{F}_K$ its special fiber. Then $\mathcal{X}$ is normal if and only if the following two conditions hold:

- (a) $\mathcal{O}_{\mathcal{X}, \eta}$ is a discrete valuation ring for every generic point $\eta \in \mathcal{X}_s$, and
- (b) $\mathcal{X}_s$ has no embedded points,

where an embedded point is a non-generic point $x \in \mathcal{X}_s$ satisfying $\mathfrak{m}_x = \text{Ann}_R(r)$ for some $r \in R := \mathcal{O}_{\mathcal{X}_s, x}$.

Proof. According to Serre's criterion for normality (cf. Theorem A.4) We need to check the properties (R₁) and (S₂), that is

- (a') $\mathcal{X}$ is regular at x when $\text{codim } x \leq 1$, and
- (b') $\text{depth } \mathcal{O}_{\mathcal{X}, x} \geq 2$

for all $x \in \mathcal{X}$. Points in $\mathcal{X}$ with codimension 1 are

1. points on the generic fibre, which is regular by assumption, and
2. the generic point on the special fibre.

Since a DVR is a regular local Noetherian domain of dimension 1 and vice versa, (a') is equivalent to (a).

According to [Liu02, Proposition 8.2.11] condition (b') is equivalent to say that $\mathcal{X}_s$ verifies $\text{depth } \mathcal{O}_{\mathcal{X}_s, x} \geq 1$. This is always true for a generic point. If x is not a generic point, it is equivalent to say that x is not an embedded point. $\square$

Example 1.4. Let us call an $\mathcal{O}_K$ -scheme a *pre-model* of X by dropping the condition of being normal. Consider a pre-model with special fibre $\mathcal{X}_s := \text{Spec } \mathbb{F}_K[x, y]/(x^2, xy)$, which is topologically

equivalent to $\mathbb{A}_{\mathbb{F}_K}^1$. We have that the point corresponding to the maximal ideal (x, y) is an embedded point.

An example with non-singular generic fiber but embedded point in special fiber can be found in [Har77, Example III.9.8.4].

Proposition 1.5 ([AW12, Proposition 2.3]). *If $\mathcal{X}_s$ is reduced, then $\mathcal{X}$ is permanent.*

Proof. Let L/K be a finite extension. Since $\mathbb{F}_K$ is assumed perfect, we have that the special fibre of $\mathcal{X}' := \mathcal{X}_{\mathcal{O}_L} = \mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L$ is $\mathcal{X}'_s := \mathcal{X}_s \otimes_{\mathbb{F}_K} \mathbb{F}_L$, which is reduced. That means $\mathcal{X}'_s$ satisfies Condition (b) in Lemma 1.3.

Claim. $\mathcal{O}_{\mathcal{X}', \eta}$ is a DVR for any $\eta \in \mathcal{X}'_s$ a generic point.

Proof. We have that $\mathcal{O}_{\mathcal{X}', \eta}$ is a Noetherian local ring of dimension 1. Meanwhile we have

$$\mathcal{O}_{\mathcal{X}'_s, \eta} = \mathcal{O}_{\mathcal{X}', \eta} / \pi_L \mathcal{O}_{\mathcal{X}', \eta}.$$

Previous argument on $\mathcal{X}'_s$ being reduced implies that $\pi_L \mathcal{O}_{\mathcal{X}', \eta}$ is a maximal ideal, hence a DVR. $\blacksquare$
The claim is exact the Condition (a) for $\mathcal{X}_{\mathcal{O}_L}$. Thus we have that $\mathcal{X}_{\mathcal{O}_L}$ is normal.

Alternative proof without using Serre's criterion for normality: [Liu02, Lemma 4.1.18]. $\square$

Proposition 1.6 ([AW12, Corollary 2.2.(ii)]). *Let $\mathcal{X}$ be a model of X . The following statements are equivalent:*

- (a) $\mathcal{X}_s$ is reduced.
- (b) $\mathcal{O}_{\mathcal{X}_s, \eta}$ is a field for every generic point $\eta \in \mathcal{X}_s$
- (c) v_Z is weakly ramified, i.e. $[\Gamma_{v_Z} : \Gamma_K] = 1$, for all $v_Z \in V(\mathcal{X})$.

Proof. Since $\mathcal{X}$ is normal, there is no embedded points in $\mathcal{X}_s$ (cf. Lemma 1.3). Since $\mathcal{X}_s$ has dimension one, which means that $\mathcal{X}_s$ is reduced if and only if $\mathcal{X}_s$ is reduced in codimension zero, i.e. $\mathcal{O}_{\mathcal{X}_s, \eta}$ is a field for every generic point $\eta \in \mathcal{X}_s$.

According to the Proposition given by the previous talk, which states that

$$[\Gamma_{v_Z} : \Gamma_K] = \text{length}(\mathcal{O}_{\mathcal{X}_s, \eta}) \quad \text{where } Z = \overline{\{\eta\}}.$$

This implies that $\mathcal{O}_{\mathcal{X}_s, \eta}$ is a field if and only if v_Z is weakly ramified for all $v_Z \in V(\mathcal{X})$. $\square$

Let us see an example to illustrate the result.

Example 1.7 (Curve with reduced special fibre). Let $K = \mathbb{Q}_3$ be the ground field. Consider the curve

$$X : y^3 = 3x^2 + 1$$

and the equation is already defined on $\mathcal{O}_K$, that means we can define a model with the same equation. $\mathcal{X}_s/\mathbb{F}_3$ is given by $0 = \bar{y}^3 - 1 = (\bar{y} - 1)^3$ which means $\mathcal{X}_s$ is not reduced.

The irreducible component is given by $y - 1$. Denote v the valuation corresponding the irreducible component. Write $z := y - 1$ and we have

$$\begin{aligned} 3x^2 &= y^3 - 1 \\ &= (z + 1)^3 - 1 \\ &= z^3 + 3z^2 + 3z, \end{aligned}$$

which is an Eisenstein equation in the right hand side. This results $v(z) = \frac{1}{3}v_G(3x^2) = \frac{1}{3} \notin \mathbb{Z} = \Gamma_K$.

Let us extend the ground field to $L = \mathbb{Q}_3[\pi]$ where $\pi^3 - 3 = 0$ and apply a coordinate transformation $(x, y) \mapsto (x_1, 1 + \pi y_1)$. We obtain a birational curve

$$X_1 : y_1^3 + \pi^2 y_1^2 + \pi y_1 = x_1^2.$$

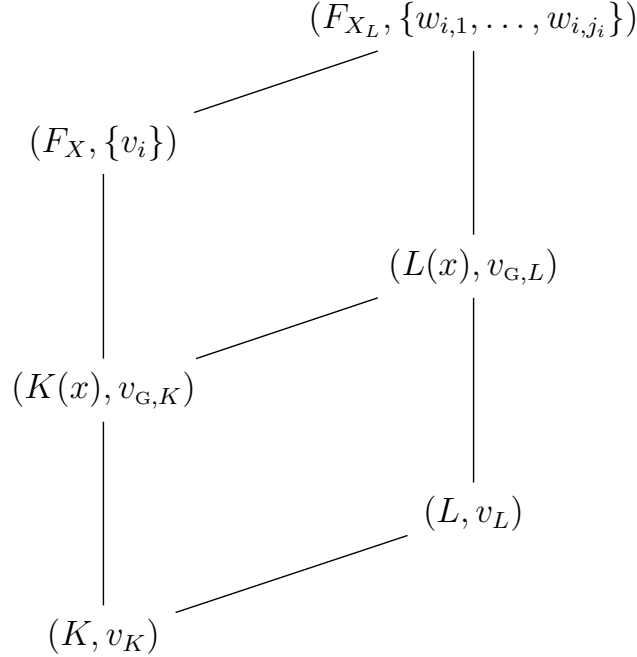
We have $y_1^3 \equiv x_1^2 \pmod{\pi}$ which means that $\mathcal{X}_s$ is reduced. $v_G(\pi y_1) = v_G(y - 1) = 1/3 = v_L(\pi) \in \Gamma_L$, especially $v_G(y_1) = 0$.

2 Model under base field extension

Let L/K be a finite field extension. Let $\mathcal{X}$ be a model of X . Denote $\mathcal{X}' := (\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L)^\sim$ the normalized base change of $\mathcal{X}$ to $\mathcal{O}_L$ and $\mathcal{X}'_s$ the special fibre of $\mathcal{X}'$. Let $\eta \in \mathcal{X}_s$ be a generic point. Let $\eta' \in \mathcal{X}'_s$ be a generic point lying over η . Since $\mathcal{X}$ (resp. $\mathcal{X}'$) is normal, we have that $R := \mathcal{O}_{\mathcal{X},\eta}$ (resp. $S := \mathcal{O}_{\mathcal{X}',\eta'}$) is a DVR dominating $\mathcal{O}_K$ (resp. $\mathcal{O}_L$) by Lemma 1.3.(a), that is

- $R \supseteq \mathcal{O}_K$ (resp. $S \supseteq \mathcal{O}_L$), and
- $\mathfrak{m}_R \cap \mathcal{O}_K = \pi_K \mathcal{O}_K$ (resp. $\mathfrak{m}_S \cap \mathcal{O}_L = \pi_L \mathcal{O}_L$).

We arrive at the following diagram of fields.



Where different v_i 's represent different valuation represented by different components of $\mathcal{X}_s$, i.e. there is a one-to-one correspondence between the generic points of $\mathcal{X}_s$ and the induced valuation on F_X (cf. Lemma 2.2). The extension on the left part is constructed artificial. The transcendental element of the field extension $K(x)/K$ can be chosen, so that the Gauß-valuation $v_{G,K}$ on $K(x)$ is compatibel in the extensions of valuations.

$w_{i,j}$ represent the valuation corresponding the irreducible component of $\mathcal{X}'_s$ whose generic point lying over the generic point of $Z \subseteq \mathcal{X}_s$, which correspondes v_i . Because of the construction on $v_{G,K}$ we have a compatibel diagram on valuations.

Theorem 2.1 (Existence of model with special fibre, [AW12, Proposition 2.3]). *Let $\mathcal{X}$ be a model of X . Then there is a finite extension L/K such that $\mathcal{X}' = (\mathcal{X} \otimes_{\mathcal{O}_K} \mathcal{O}_L)^\sim$ has a reduced special fibre.*

Proof. By [Epp73, Theorem 1.0] there exists a field extension L/K such that $w_{i,j}$ is weakly unramified over v_L . According to Proposition 1.6 $\mathcal{X}'_s$, the special fibre of $\mathcal{X}'$, is reduced. $\square$

In the proof we have used the Theorem of Epp. Let us have a look on the statement in case of tamely ramified extension.

Lemma 2.2 (Abhyanka's lemma). *Let L/K be a finite extension. Let $\{v_1, \dots, v_n\} = V(X)$ denote the extended valuation on the function field F_X/K . We have that $F_{X_L} = F_X \cdot L$ and with the extended valuation denoted as $w_{1,1}, \dots, w_{n,j_n}$ respectively. Assume that $v_i/v_{G,K}$ are tamely ramified for all i . If $e_i := e(v_i/v_{G,K})$ divides $e(v_{G,L}/v_{G,K}) = e(v_L/v_K)$ for all i , then $w_{i,j} \in V(X_L)$ is weakly unramified over $v_{G,L}$ for all i, j .*

Proof. We will use the equivalence of Proposition 1.6 without further mention, that is the equivalence between reduced special fibre and weakly unramified valuation. W.l.o.g. we may assume the followings:

- $n = 1$ and $j_1 = 1$ since the argument can be constructed inductively,
- $e_1 =: e = e(v_1/v_{G,K}) = e(v_{G,L}/v_{G,K})$ by the assumption of perfect residue field, and
- K contains the e -th root of unity, i.e. $\pi_K = \pi_L^e$, since we can always operate a Kummer extension on K without violating other assumptions and obtain a compatible diagram.

Since $F_{X_L} = F_X \cdot L$ we have that $[\Gamma_{w_1} : \Gamma_{v_1}] = 1$. By multiplicativity of ramification indices we conclude $[\Gamma_{w_1} : \Gamma_{v_L}] = 1$ $\square$

With Abhyanka's lemma we know about necessary field extension for a model become permanent if the v_i 's are tamely ramified. It can be defined directly and uniquely when $\mathbb{F}_K$ is algebraically closed.

Example 2.3 (Reduction after wild ramified field extension). Let us consider the curve over $K := \mathbb{Q}_2$

$$X : y^2 = 8x^3 - 2.$$

1. First let us consider the field extension $L_1 = K(\pi_1)$ where $\pi_1^2 - 2 = 0$ and the coordinate transformation $(x, y) \mapsto (x_1, 2y_1 - \pi_1 - 2)$. We obtain a birational curve

$$X_1 : y_1^2 - (\pi_1 + 4)y_1 = 2x_1^3 - (2 + \pi_1)$$

with $v_{G,L_1}(y_1) = \frac{1}{4}$. We have $[\Gamma_{F_{X_1}} : \Gamma_{L_1}] = 2 \neq 1$.

2. let us consider the field extension $L_2 = K(\pi_2)$ where $\pi_2^2 + 2 = 0$ and the coordinate transformation $(x, y) \mapsto (x_2, 2\pi_2 y_2 + \pi_2)$. We obtain a birational curve

$$X_2 : y_2^2 + y_2 = -x_2^3$$

which is given by an equation of AS-Form, i.e. its special fibre is a reduced curve. That means $[\Gamma_{F_{X_2}} : \Gamma_{L_2}] = 1$.

Since a wild ramified field extension is not unique, the choice of right extension is important for having reduced special fibre/permanent model.

A Serre's criterion for normality

Definition A.1. Let $(R, \mathfrak{m})$ be a Noetherian local ring. Let M be an R -module.

1. An element $r \in R$ is M -regular if the multiplication-by- r -map $m_r : M \rightarrow M$ is injective.
2. A sequence of elements $r_1, \dots, r_n \in R$ is called an M -regular sequence if

$$r_i \text{ is } (M/(r_0M + \dots + r_{i-1}M))\text{-regular for all } 1 \leq i \leq n$$

(with convention $r_0 = 0$).

3. The *depth* of M , denote as $\text{depth}(M)$, is the maximal number of elements of an M -regular sequence with elements in $\mathfrak{m}$.

Remark A.2. Let $(R, \mathfrak{m})$ be a Noetherian local ring. For $\mathfrak{p} \in \text{Spec } R$ we have

$$\text{depth } \mathfrak{p} = 0 \iff \mathfrak{p} = \text{Ann}(r) \text{ for some } r \in R.$$

Definition A.3. Let X be a locally Noetherian scheme and $k \in \mathbb{N}_0$. We say that

1. X verifies property (R_k) , if X is regular at all of its points of codimension smaller or equal to k , and
2. X verifies property (S_k) , if for any point $x \in X$, we have

$$\text{depth}(\mathcal{O}_{X,x}) \geq \inf\{k, \dim \mathcal{O}_{X,x}\}.$$

Theorem A.4. (*Serre's criterion for normality*) Let X be a locally Noetherian connected scheme. Then X is normal if and only if it verifies properties (S_2) and (R_1) .

References

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